Projectile motion

AS Level | 9702 AS Ch.2 | Chapter 2: Accelerated motion | Lesson 2.13 | Sections 2.13

Goal: Treat horizontal and vertical motion separately for an object launched horizontally.

velocity

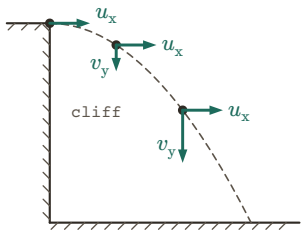
01 The big picture

With no air resistance, a projectile's horizontal and vertical motions are independent. Horizontally there is no force, so the velocity stays constant. Vertically it accelerates at g downwards, exactly like a dropped object.

The time in the air is the link: find it from the vertical motion, then use it for the horizontal distance. The path is a parabola.

Projectile [paraphrase]
An object moving under gravity alone after it is launched.

Fig. 1 · Thrown horizontally from a cliff



The horizontal arrows stay the same length; the vertical arrows grow by g every second.

02 Key equations

Horizontal (constant)
$$x = u_x t$$

Vertical (from rest)
$$y = \tfrac{1}{2} g t^2$$

Vertical velocity
$$v_y = g t$$

Symbols and units

x, y	horizontal, vertical distance	m	u_x	horizontal velocity	$\mathrm{m\,s^{-1}}$	v_y	vertical velocity	$\mathrm{m\,s^{-1}}$

03 Sample problems with solutions

Sample 1 · Rolling off a table

A ball rolls off a table 1.8 m high at 3.0 m s⁻¹. Calculate how far from the table it lands.

Step 1. Vertical: falls 1.8 m from rest:
$$t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2 \times 1.8}{9.81}} = 0.606\ \mathrm{s}$$

Step 2. Horizontal: constant 3.0 m s⁻¹:
$$x = u_x t = 3.0 \times 0.606 = 1.82\ \mathrm{m}$$

Answer: $x = 1.8$ m from the table

Unit check: $\mathrm{m\,s^{-1}} \times \mathrm{s} = \mathrm{m}$ ✓

04 Practice problems

P1 A ball is dropped as an identical ball is thrown horizontally from the same height. Which lands first?
Answer: They land together

P2 A stone is thrown horizontally at 15 m s⁻¹ from a 45 m cliff (Fig. 1). How long is it in the air?
Answer: 3.0 s (3.03 s)

P3 How far from the cliff does it land?
Answer: 45 m (45.4 m)

P4 Find its vertical velocity just before it lands.
Answer: 30 m s⁻¹ (29.7 m s⁻¹) downwards

05 Other useful information

Common mistakes

✗ **Wrong:** Let the horizontal velocity change during the flight.
✓ **Right:** With no air resistance it stays u_x; only v_y grows.

Exam tips

- Make two columns, horizontal and vertical; only t crosses between them.

What the syllabus asks

- 9702 topic 2: uniform velocity plus a perpendicular uniform acceleration. Coursebook 2.13.

Explore: Practice questions for this lesson